

CARDIOVIZ XR

48-Hour Medical Animation Challenge — 1-Page Creative Breakdown

1. Creative Concept

Concept: “From Silent Damage to Sudden Blockage.” A cinematic vessel cutaway makes an invisible cardiovascular process understandable to a non-medical viewer. Distinct visual cues identify endothelial injury, ApoB particles, plaque, narrowing and thrombus formation.

2. Narrative Flow

Stage	Visual Story	Comprehension Goal
01 — Endothelial injury	Healthy arterial lining develops localized micro-damage/shearing.	Show the first invisible change.
02 — ApoB retention	ApoB-containing particles become retained in the vessel wall; early plaque develops.	Show how accumulation begins.
03 — Plaque growth	Plaque enlarges and the arterial lumen visibly narrows.	Make progression and loss of space obvious.
04 — Rupture + clot	Unstable plaque ruptures and a thrombus rapidly forms at the rupture site.	Explain the final acute event.

3. Lighting & Texturing

Lighting: Soft clinical key light with controlled rim lighting reveals vessel geometry; localized highlights emphasize plaque growth and rupture. **Materials:** Semi-translucent arterial tissue with subtle wet/specular response, with clearly differentiated particle, plaque and clot materials. The look prioritizes medical clarity over unnecessary photorealism.

4. Animation & Pacing

The camera progresses from a wider vessel view into closer cross-sectional detail as disease advances. Transitions remain continuous: injury → retention → plaque growth → rupture → clot. Short stage labels are used only where they improve comprehension.

5. Asset & Production Notes

Toolchain: [Insert actual tool(s) used]. **Original work:** Vessel, plaque, particles, clot, camera, lighting, animation and compositing created for this challenge. **Third-party assets:** [List any stock/HDRI/plugin, or “None”].

6. Generative-AI Disclosure

AI use: [Select the truthful statement.] “No generative AI was used in the final animation.” OR “Generative AI was used for [specific task]. It was not used to replace the core 3D animation work.” Disclose all AI-assisted elements as required.

7. Patient-Comprehension Goal

After one viewing, a non-medical viewer should understand the progression from healthy vessel → endothelial damage → ApoB/plaque accumulation → narrowing → plaque rupture → thrombus formation.